# Giovanni Luca Marchetti

**Researcher in Deep Learning** | Email: glma@kth.se | Webpage
AI4S, Stockholm, Sweden

## WORK EXPERIENCE

- Senior Scientist and Consultant — 06/2026 – Now
  *AI4S AB, Stockholm, Sweden*
  - Working alongside Swedish universities on deep learning projects across natural and social sciences.
- Postdoctoral Researcher — 05/2024 – 06/2026
  *Royal Institute of Technology (KTH), Stockholm, Sweden*
  - Worked on theoretical approaches to deep learning via algebraic geometry.
  - *Supervisor*: Kathlén Kohn.
- Research Intern — 06/2023 – 10/2023
  *Qualcomm AI Research, Amsterdam, The Netherlands*
  - Worked on geometric deep learning for combinatorial optimization, with applications to signal processing.

## EDUCATION

- Ph.D. Computer Science — 05/2020 – 04/2024
  *Royal Institute of Technology (KTH), Stockholm, Sweden*
  - Worked on geometric deep learning and high-dimensional statistics, with applications to robotics.
  - *Thesis title*: On Symmetries and Metrics in Geometric Inference.
  - *Supervisor*: Danica Kragic. *Co-supervisor*: Anastasia Varava.
- Ph.D. (uncompleted) Pure Mathematics — 10/2017 – 03/2020
  *University of Sheffield, United Kingdom*
  - Worked on algebraic geometry and category theory, with applications to string theory.
  - *Supervisor*: Tom Bridgeland.
- M.Sc. Pure Mathematics — 10/2015 – 07/2017
  *University of Rome La Sapienza, Italy*
  - *Grade*: Summa cum laude, with an excellence award.
- B.Sc. Pure Mathematics — 10/2012 – 07/2015
  *University of Rome La Sapienza, Italy*
  - *Grade*: Summa cum laude, with an excellence award.

## PUBLICATIONS

- Giovanni Luca Marchetti†, Erin Connelly†, Paul Breiding†, and Kathlén Kohn†. *Critical Points of Degenerate Metrics on Algebraic Varieties: A Tale of Overparametrization*. In: Preprint. 2026.
- Vahid Shahverdi†, Giovanni Luca Marchetti†, Georg Bökman†, and Kathlén Kohn†. *Identifiable Equivariant Networks are Layerwise Equivariant*. In: International Conference on Machine Learning (ICML). 2026
- Giovanni Luca Marchetti†, Daniel Kunin†, Adele Myers, Francisco Acosta, and Nina Miolane. *Sequential Group Composition: A Window into the Mechanics of Deep Learning*. In: International Conference on Machine Learning (ICML). 2026.
- Vahid Shahverdi†, Giovanni Luca Marchetti†, and Kathlén Kohn†. *Learning on a Razor's Edge: Identifiability and Singularity of Polynomial Neural Networks*. In: International Conference on Learning Representations (ICLR). 2026.
- Aniss Aiman Medbouhi†, Alejandro García-Castellanos†, Giovanni Luca Marchetti, Daniel Pelt, Erik J Bekkers, and Danica Kragic. *Randomized HyperSteiner: A Stochastic Delaunay Triangulation Heuristic for the Hyperbolic Steiner Minimal Tree*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2026.
- Daniel Kunin, Giovanni Luca Marchetti, Feng Chen, Dhruva Karkada, James B Simon, Michael R DeWeese, Surya Ganguli, and Nina Miolane. *Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks*. In: Advances in Neural Information Processing Systems (NeurIPS). 2025.
- Zheyu Zhuang, Ruiyu Wang, Giovanni Luca Marchetti, Florian T Pokorny, and Danica Kragic. *MirrorDuo: Reflection-Consistent Visuomotor Learning from Mirrored Demonstration Pairs*. In: Annual Conference on Robot Learning (CoRL). 2025.

- Giovanni Luca Marchetti†, Vahid Shahverdi†, Stefano Mereta†, Matthew Trager†, and Kathlén Kohn†. *Algebra Unveils Deep Learning: An Invitation to Neuroalgebraic Geometry*. In: International Conference on Machine Learning (ICML). 2025. **Spotlight Paper**.
- Nathan W Henry†, Giovanni Luca Marchetti†, and Kathlén Kohn†. *Geometry of Lightning Self-Attention: Identifiability and Dimension*. In: International Conference on Learning Representations (ICLR). 2025.
- Vahid Shahverdi†, Giovanni Luca Marchetti†, and Kathlén Kohn†. *On the Geometry and Optimization of Polynomial Convolutional Networks*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2025.
- Giovanni Luca Marchetti, Gabriele Cesa, Kumar Pratik, and Arash Behboodi. *Neural Lattice Reduction: A Self-Supervised Geometric Deep Learning Approach*. In: Transactions on Machine Learning Research (TMLR), 2025.
- Alejandro García-Castellanos, Aniss Aiman Medbouhi, Giovanni Luca Marchetti, Erik J Bekkers, and Danica Kragic. *Hypersteiner: Computing Heuristic Hyperbolic Steiner Minimal Trees*. In: Symposium on Algorithm Engineering and Experiments (ALENEX), 2025.

- Giovanni Luca Marchetti, Christopher J Hillar, Danica Kragic, and Sophia Sanborn. *Harmonics of Learning: Universal Fourier Features Emerge in Invariant Networks*. In: Conference on Learning Theory (COLT). 2024.
- Aniss Aiman Medbouhi, Giovanni Luca Marchetti, Vladislav Polianskii, Alexander Kravberg, Petra Poklukar, Anastasia Varava, and Danica Kragic. *Hyperbolic Delaunay Geometric Alignment*. In: European Conference on Machine Learning (ECML-PKDD). 2024.
- Giovanni Luca Marchetti, Vladislav Polianskii, Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *An Efficient and Continuous Voronoi Density Estimator*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2023. **Notable Paper award**.
- Giovanni Luca Marchetti†, Gustaf Tegnér†, Anastasiia Varava, and Danica Kragic. *Equivariant Representation Learning via Class-Pose Decomposition*. In: International Conference on Artificial Intelligence and Statistics (AISTATS). 2023.
- Alfredo Reichlin†, Giovanni Luca Marchetti†, Hang Yin, Anastasiia Varava, and Danica Kragic. *Learning Geometric Representations of Objects via Interaction*. In: European Conference on Machine Learning (ECML-PKDD). 2023.
- Luis Perez Rey†, Giovanni Luca Marchetti†, Danica Kragic, Dimitri Jarnikov, and Mike Holenderski. *Equivariant Representation Learning in the Presence of Stabilizers*. In: European Conference on Machine Learning (ECML-PKDD). 2023.
- Alexander Kravberg†, Giovanni Luca Marchetti†, Vladislav Polianskii†, Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *Active Nearest Neighbor Regression Through Delaunay Refinement*. In: International Conference on Machine Learning (ICML). 2022.
- Vladislav Polianskii†, Giovanni Luca Marchetti†, Alexander Kravberg, Anastasiia Varava, Florian T Pokorny, and Danica Kragic. *Voronoi Density Estimator for High-Dimensional Data: Computation, Compactification and Convergence*. In: Uncertainty in Artificial Intelligence (UAI). 2022.
- Alfredo Reichlin, Giovanni Luca Marchetti, Hang Yin, Ali Ghadirzadeh, and Danica Kragic. *Back to the Manifold: Recovering from Out-of-Distribution States*. In: International Conference on Intelligent Robots and Systems (IROS). 2022.
- Domenico Fiorenza†, Fosco Loregian†, and Giovanni Luca Marchetti†. *Hearts and Towers in Stable Infinity-Categories*. In: Journal of Homotopy and Related Structures. 2019.

*Note: The symbol † denotes shared first-authorship.*
A complete list of publications is available on Google Scholar [LINK].

## PATENTS

- Giovanni Luca Marchetti, Cesa Gabriele, Kumar Pratik, and Arash Behboodi. *Geometric Deep Learning for Lattice Reduction*. US Patent US12413270B2. 2024.

## TEACHING EXPERIENCE

Contributed to the following courses:

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|---|---|
| · Geometry and Topology for Data Analysis (guest lectures) | Univ. of Trento, 2025 |
| · Groups and Rings (guest lectures) | KTH, 2025 |
| · Advanced Linear Algebra (tutorials, examination) | KTH, 2025 |
| · Artificial Intelligence (tutorials, examination, organization) | KTH, 2022 – 2023 |
| · Computer Vision (tutorials, examination) | KTH, 2023 |
| · Database Technology (tutorials, examination) | KTH, 2021 – 2022 |
| · Scientific Programming in Python (tutorials) | Univ. of Sheffield, 2019 |
| · Mechanics and Fluids (tutorials, examination) | Univ. of Sheffield, 2019 |
| · Analysis and Algebra (tutorials) | Univ. of Sheffield, 2018 – 2019 |

## SUPERVISORY AND ORGANIZATIONAL ACTIVITY

Mentored the following students:

- Juan Sepulveda (Ms.Sc., supervision) 2026
- Johan Slettengren (Ms.Sc., supervision) 2025
- Aniss Medbouhi (Ph.D., co-supervision) 2023 – Now
- Nathan Henry (research visit, co-supervision) 2024
- Alejandro García Castellanos (Ms.Sc. and research engineer, co-supervision) 2023 – 2024
- Markus Hector (Ms.Sc., supervision) 2023

Co-organized the following events and activities:

- Workshop: *Geometric Deep Learning in Umeå (GeUmetric)* 2025
- Reading Seminar: *Computer Vision and Deep Learning* 2022 – 2023
- Reading Seminar: *Geometry and Machine Learning* 2021 – 2023

## SELECTED INVITED TALKS

- *The Algebraic Geometry of Deep Learning* — Lausanne (Switzerland), 2025
  EPFL AI Center Center Seminar
- *An Invitation to Neuroalgebraic Geometry* — Madison (USA), 2025
  SIAM Conference on Applied Algebraic Geometry (AG)
- *On the Geometry and Optimization of Polynomial Convolutional Network* — Seattle (USA), 2025
  Joint Meeting of Mathematicians (JMM)
- *Harmonics of Learning: Universal Fourier Features Emerge in Invariant Networks* — Edmonton (Canada), 2024
  International Conference on Learning Theory (COLT)
- *Equivariant Representation Learning in the Presence of Stabilizers* — Turin (Italy), 2023
  European Conference on Machine Learning (ECML-PKDD)
- *An Efficient and Continuous Voronoi Density Estimator* — Valencia (Spain), 2023
  International Conference on Artificial Intelligence and Statistics (AISTATS)
- *Active Nearest Neighbor Regression Through Delaunay Refinement* — Baltimore (USA), 2022
  International Conference on Machine Learning (ICML)
- *Higher Categories in Mirror Symmetry* — Univ. of Warwick (UK), 2019
  Junior Algebraic Geometry Seminar

## SELECTED SOFTWARE

Developed the following machine learning software:

- [LINK] A deep self-supervised framework for learning group-equivariant representations (in PyTorch).
- [LINK] A contrastive complex-valued network based on harmonic analysis (in PyTorch, JAX).
- [LINK] A high-dimensional non-parametric density estimator based on Voronoi cells (in `C++`, NumPy).

Further software is available on GitHub [LINK].

## SKILLS

- *Languages*: Italian (native), Russian (native), English (professional), Swedish (intermediate), French (basic).
- *Programming*: `Python` (NumPy, PyTorch, JAX), `C`, `C++`, LaTeX, `Bash`, SQL.

## REFERENCES

Contact information for reference requests:

- kathlen@kth.se Kathlén Kohn, Associate Prof. at KTH
- dani@kth.se Danica Kragic, Full Prof. at KTH
- anastasia.varava@seb.se Anastasia Varava, Head of Research at SEBx